

Model Analysis Report

BRI-Adjusted Floor Level Model

MKM-BRF-001

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Governance Framework: SR 11-7 / SS1/23 Model Risk Management

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1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: Floor-level uplift (m) by BRI flood sub-score. No credit at or below 0.38, the full 3.0 m at or above 0.87, linear in between.

BRI flood score	Uplift (m)	Band
0.0000	0.0000	no credit
0.1900	0.0000	no credit
0.3000	0.0000	no credit
0.3800	0.0000	no credit
0.5000	0.7350	linear ramp
0.6200	1.47	linear ramp
0.7450	2.23	linear ramp
0.8700	3.00	full credit
0.9350	3.00	full credit
1.00	3.00	full credit

Table 2: Adjusted floor level (m) by surveyed floor level and BRI grade mid-point score. The adjustment is additive, so the adjusted level never falls below the surveyed level.

Surveyed floor (m)	NR (0.190)	B (0.500)	A (0.745)	AA (0.935)
0.0000	0.0000	0.7350	2.23	3.00
0.1500	0.1500	0.8850	2.38	3.15
0.3000	0.3000	1.03	2.54	3.30
0.6000	0.6000	1.33	2.83	3.60
1.00	1.00	1.74	3.23	4.00
1.50	1.50	2.23	3.73	4.50

Table 3: Letter-grade fallback: representative flood score and resulting uplift. A trailing + modifier is stripped before lookup; unknown or non-applicable grades return no representative score.

BRI rating	Representative score	Uplift (m)
AA	0.9350	3.00
AA+	0.9350	3.00
A	0.7450	2.23
A+	0.7450	2.23
B	0.5000	0.7350
NR	0.1900	0.0000
N/A	not mappable	n/a
ZZ	not mappable	n/a

Table 4: Flood trigger by BRI grade for a property with a 0.3 m surveyed floor level. The credit raises the depth at which the property first floods, which is how the model suppresses PRS flood event counts.

Water above ground (m)	NR grade	B grade	A grade	AA grade
0.2500	no	no	no	no
0.5000	yes	no	no	no
1.00	yes	no	no	no
1.50	yes	yes	no	no
2.00	yes	yes	no	no
3.00	yes	yes	yes	no
4.00	yes	yes	yes	yes

Table 5: Damage ratio through the baseline depth-damage curve, evaluated at the depth above the adjusted floor level. The gap between NR and AA at a given water level is the resilience credit expressed as avoided loss.

Water above ground (m)	NR grade	B grade	A grade	AA grade
0.2500	0.0000	0.0000	0.0000	0.0000
0.5000	0.1167	0.0000	0.0000	0.0000
1.00	0.3100	0.0000	0.0000	0.0000
1.50	0.4400	0.2346	0.0000	0.0000
2.00	0.5400	0.3896	0.0000	0.0000
3.00	0.7050	0.5931	0.2346	0.0000
4.00	0.8200	0.7448	0.4931	0.3100

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 6: Stress testing scenarios for BRI-Adjusted Floor Level Model.

Scenario	Parameter Shock	Severity	Status
Baseline	No shock	—	Passed
Moderate stress	+2 σ inputs	Medium	Pending
Severe stress	+3 σ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model's validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 7: Performance metrics for BRI-Adjusted Floor Level Model.

Metric	Value	Status
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.